Performance Vocabulary

Latency
Average time per requests

Throughput
Completed requests per second

Tail Outlier
95 Percentile and 99 Percentile expose outliers because throughput average hides the worst experiences

Concurrency
Requests sent concurrently (a.k.a "in flight") at one time per client

Where to optimize?

Problem

Speeding up a non-bottleneck may produce negligible benefit

Example

A backend can reply in 20 ms but requires 700 ms in average for a database request

Solution

Break latency into “spans” instead of guessing